

SOME NOTES ON LP AND OPTIMAL TRANSPORT

GIULIO TRIGILA

ABSTRACT. This is a set of informal notes introducing to linear programming (LP) and showing how this is connected to the discrete Kantorovich problem in optimal transport. There is nothing new here, the LP part is taken from the Linear Algebra book by Peter Lax. Please let me know if you think something is not making sense and feel free to suggest improvements to this document.

1. LINEAR PROGRAMMING

In this section we will review the fundamentals of the theory of duality in linear programming. We will start showing how the theorem of the alternative for linear systems is generalized when inequality constraints are added. This will be useful to prove the duality principle. We will proof only the main theorems referring to the literature when a proof is not provided.

Every vector is intended to be a column vector, while row vectors are marked with an apex. Apices are also used to indicate the transpose of a matrix.

2. THEOREM OF THE ALTERNATIVE

Let $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$ and $b \in \mathbb{R}^m$. The solvability of the linear system $Ax = b$ is characterized by the following theorem

Theorem 2.1. (*Fredholm*). *Either the linear system $Ax = b$ has a solution or exists v such that $v'A = 0$ and $v'b \neq 0$.*

In words either b is in the column space of A or otherwise b has a nonzero component in the null space of A' .

The theorem (2.1) can be generalized in the case we are requiring the solution of $Ax = b$ to be (component-wise) non negative

Theorem 2.2. (*Farkas*). *Either the problem*

$$(2.3) \quad \begin{cases} Ax = b \\ x \geq 0 \end{cases}$$

has a solution or there exists a v such that $v'A \geq 0$ and $v'b < 0$.

The details of the proof can be found in [?], here we will just give the main idea of why the theorem holds.

Let K be the set of vectors obtained by a linear combination, with non negative coefficients, of the columns of A . This set can be represented as an open-ended cone (with its vertex in the origin) and it can be proved that K is convex and closed. These two condition on the set K , are sufficient for the following theorem to hold:

Theorem 2.4. (*Hyperplane separation*). *Let S be a closed, convex set and y a point not in S . Then there is a closed half-space that contains S but not y .*

Because of theorem (2.4) either b is in K , namely system (2.3) has a solution, or otherwise it is possibile to find an hyperplane that goes through the origin and that has the vector b on one side and the set K on the other. Let v be the vector perpendicular to such plane. Then the angle in between v and any vector in K will be less or equal to $\pi/2$, namely $v'A \geq 0$ while the angle in between v and b is going to be bigger than $\pi/2$ resulting in $v'b < 0$.

Theorem (2.2) can be easily generalized to the following case

Theorem 2.5. *Either the problem*

$$(2.6) \quad \begin{cases} Ax \leq b \\ x \geq 0 \end{cases}$$

has a solution or there exists a vector $v \geq 0$ such that $v'A \geq 0$ and $v'b < 0$.

Proof. It is sufficient to notice that the problem (2.5) can be recasted in the form of (2.2) by using a *slack variable* z . By defining

$$(2.7) \quad z = b - Ax, \quad \tilde{x} = \begin{bmatrix} z \\ x \end{bmatrix} \quad \text{and} \quad \tilde{A} = [I \quad A]$$

we can rewrite (2.3) as

$$(2.8) \quad \begin{cases} \tilde{A}\tilde{x} = b \\ \tilde{x} \geq 0 \end{cases}$$

The equations in (2.8) have the same form of those one in (2.3) so we can use the theorem (2.2) to assert that either (2.8) has a solution or there exists v s.t. $v'\tilde{A} \geq 0$ and $v'b < 0$. This is equivalent to the statement of the theorem. $\square$

3. DUALITY

In this section we will use theorem (2.5) to prove the principle of duality and then the minimax theorem. The principle of duality concerns with the two following optimization problems:

$$(3.1) \quad \begin{cases} s = \inf_x c'x \\ Ax \geq b \\ x \geq 0 \end{cases} \quad \text{and} \quad \begin{cases} S = \sup_y y'b \\ y'A \leq c \\ y \geq 0 \end{cases}$$

where $A \in \mathbb{R}^{m \times n}$, x and $c \in \mathbb{R}^n$ and $y, b \in \mathbb{R}^m$. We will call the vector x feasible if the constraints $Ax \geq b$ and $x \geq 0$ are satisfied. Similarly y is defined to be feasible if $y'A \leq c$ and $y \geq 0$ are satisfied.

Theorem 3.2. (*Duality*). Suppose there exist feasible vectors x and y . Then S and s are finite and $S = s$.

Proof. Since x must be a vector with non negative components then by multiplying both sides the first inequality constraint for the sup problem we get

$$(3.3) \quad c'x \geq y'Ax \geq y'b$$

or equivalently $s \geq S$. To prove the equality in between s and S it is sufficient to show the existence of a feasible y for the sup problem in (3.1) such that

$$(3.4) \quad y'b \geq s.$$

In order to consider a feasible y for the sup problem that at the same time satisfies eq. (3.4) we will combine the inequalities $A'y \leq c$ (equivalent to $y'A \leq c$) and $y \geq 0$ with (3.4) as follow

$$(3.5) \quad \begin{bmatrix} A' \\ -b \end{bmatrix} y \leq \begin{bmatrix} b \\ -s \end{bmatrix}, \quad y \geq 0$$

Let's show the existence of such y by contradiction. Let's suppose that there is a feasible y for the sup problem that does not satisfies (3.4). Then such y does not satisfies (3.5) and according to theorem (2.5) there exist a vector v and a scalar a such that

$$(3.6) \quad [v' \ a] \geq 0, \quad [v' \ a] \begin{bmatrix} A' \\ -b \end{bmatrix} \geq 0, \quad [v' \ a] \begin{bmatrix} b \\ -s \end{bmatrix} < 0.$$

Clearly if $a = 0$ the system (3.6) reduces to

$$(3.7) \quad v \geq 0, \quad v'A' \geq 0, \quad v'b < 0$$

meaning that y (again in virtue of theorem (2.5)) is not feasible contrary to our assumption. This means that a has to be positive rather than non negative. Without lost of generality let $a = 1$. With this choice eq. (3.6) leads to the contradiction

$$(3.8) \quad v \geq 0, \quad v'A \geq b, \quad v'b < s.$$

saying that exists a feasible vector v for the infimum problem satisfying $v'b < \inf_x c'x$. This contradiction is consequence of the fact that we supposed the existence of a feasible y not satisfying (3.4). Summarizing, we can find a feasible y satisfying (3.4) and $S \geq s$. This fact, together with (3.3), proves that $S = s$. $\square$

Remark 3.9. The minimization and maximization problem in (3.3) are often called *primal* and *dual* problem respectively.

Remark 3.10. Clearly s is finite since the entries of b , A and c are finite. The inequality (3.4) is indeed an equality since otherwise would contradict the fact that $S = s$. This says that the supremum in (3.3) is achieved, namely that the supremum is indeed a maximum. Analogously one can show that the infimum is indeed a minimum by turning the supremum into an infimum (this can be realized replacing A , c and b with $-A$, $-c$ and $-b$).

Remark 3.11. By using theorem (2.2) rather than its generalization (2.5) we can, repeating the same proof for the theorem (3.2), show that given the primal and

dual linear programming problems

$$(3.12) \quad \begin{cases} s = \min_x c'x \\ Ax = b \\ x \geq 0 \end{cases} \quad \text{and} \quad \begin{cases} S = \max_y y'b \\ c \geq y'A \end{cases}$$

we have that $S = s$

We close this section with a set of relation that must be satisfied in correspondence of the optimal solution and that goes under the name of *complementary slackness conditions*.

Theorem 3.13. (*Equilibrium theorem*). *The feasible vectors x and y are optimal if and only if the following relations are satisfied at the same time*

- If $(Ax)_i > b_i$ then $y_i = 0$
- If $(y'A)_j < c_j$ then $x_j = 0$

The proof of the theorem relies on the fact that at the optimal solution the primal and the dual problem give the same result. In particular starting from the true relation

$$(3.14) \quad yb \leq y(Ax) = (yA)x \leq cx$$

we should investigate under which conditions we have that $yb = cx$. This equality is true if and only if the solution is optimal. There is only one way in which $yb = y(Ax)$ holds: every time we have a feasible solution satisfying strictly the inequality constraint $b_i < (Ax)_i$ the factor y_i multiplying the i -th component must be zero. At the same time the only way of having $(yA)x = cx$ is that whenever we have $(yA)_j < c_j$ then the j -th component of x must be zero.

4. A PARTICULAR CASE OF LP: DISCRETE OPTIMAL TRANSPORT

In the following we will describe the discrete version of the Monge-Kantorovich problem in its primal and dual form. Let $\rho \in \mathbb{R}^m$, $\mu \in \mathbb{R}^n$ and $c \in \mathbb{R}^{m \times n}$ be characterized by nonnegative entries and by $\sum_i \rho_i = \sum_j \mu_j$.

The discrete version of the Monge Kantorovich problem is

$$(4.1) \quad \begin{cases} \min_{\pi} \sum_{i,j} c_{ij} \pi_{ij} \\ \sum_j \pi_{ij} = \rho_i \\ \sum_i \pi_{ij} = \mu_j \\ \pi_{ij} \geq 0 \end{cases}$$

where ρ and μ represents respectively the starting and the target mass densities and c is the cost matrix.

The problem in (4.1) is equivalent to the primal linear programming problem in (3.12) once we make a correspondence in between the following quantities:

$$(4.2) \quad B = \begin{pmatrix} I & 0 & \cdots & 0 \\ 0 & I & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & I \end{pmatrix}, \quad C = \begin{pmatrix} e^1 & e^1 & \cdots & e^1 \\ e^2 & e^2 & \cdots & e^2 \\ \vdots & \vdots & \ddots & \vdots \\ e^m & e^m & \cdots & e^m \end{pmatrix}, \quad A = \begin{bmatrix} B \\ C \end{bmatrix}$$

where B is a $(n \times nm)$ matrix with $I = (1, \dots, 1)$ and C is a $(m \times nm)$ matrix with $e_i^k = \delta_{i,k}$

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$$\begin{aligned}
 (4.3) \quad x &= (\pi_{11}, \pi_{12}, \dots, \pi_{1m}, \pi_{21}, \dots, \pi_{nm})' \\
 c &= (c_{11}, c_{12}, \dots, c_{1m}, c_{21}, \dots, c_{nm}) \\
 b &= (\rho_1, \dots, \rho_m, \mu_1, \dots, \mu_m)
 \end{aligned}$$

By defining

$$(4.4) \quad y = (u_1, \dots, u_n, v_1, \dots, v_m)$$

and using (4.2) and (4.2), we can formulate the dual of the problem (4.1) as

$$(4.5) \quad \begin{cases} \max_{u,v} \sum_i u_i \rho_i + \sum_j v_j \mu_j \\ u_i + v_j \leq c_{ij} \text{ for } i = 1, \dots, n; j = 1, \dots, m. \end{cases}$$